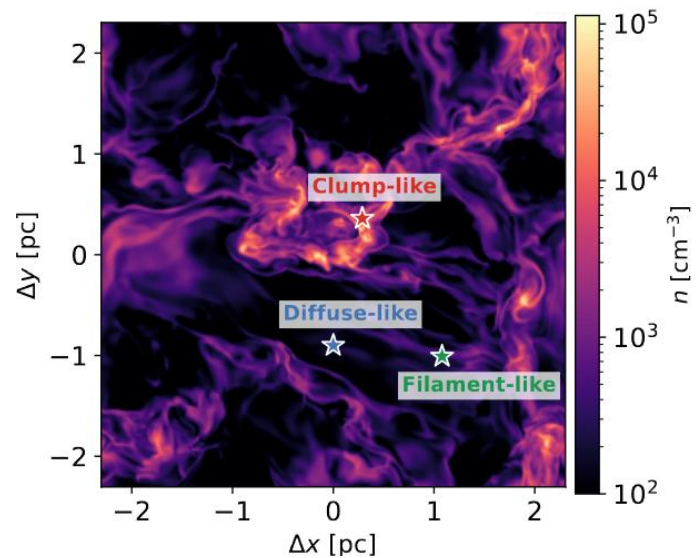
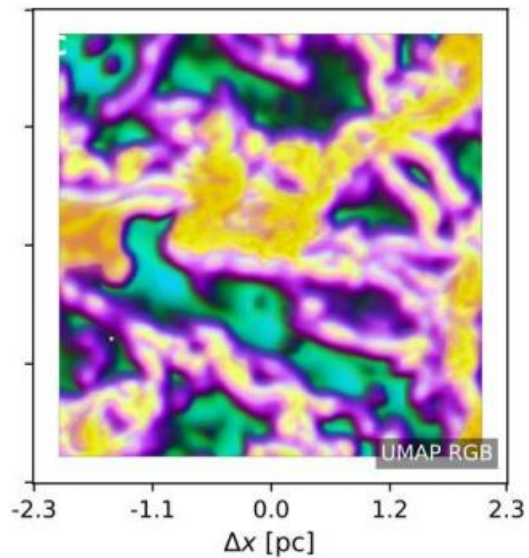


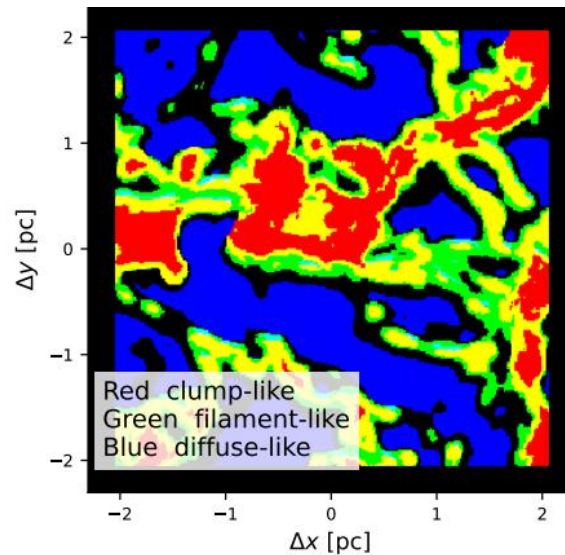
(a) Representative morphology anchors



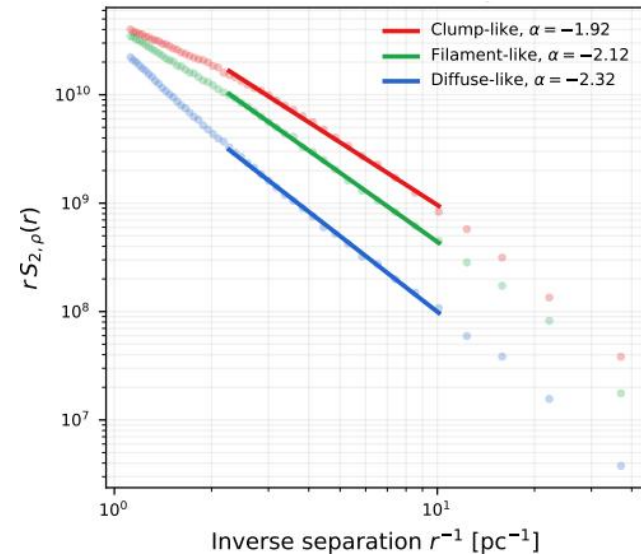
(b) Latent morphology map



(c) Latent structural neighborhoods



(d) latent-conditioned turbulent scaling



ScaleAware-JEPA
latent prediction

withheld during training:
velocity and magnetic fields

Morphology embedding
 $z(x)$

**Post-training
physical calibration**